



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer:

This program is designed to help manage and track incident tickets raised for automated bots (RPA processes) within an organization. It allows users to log new incidents, view all currently active tickets, look up a specific ticket by its ID, remove tickets once they've been resolved, and get a quick count of how many incidents are still outstanding. Overall, it functions as a simple, menu-driven ticket tracking tool that keeps an internal record of bot failures so IT or support staff can monitor and manage them from a single console interface.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

incidents.append([incident_id, bot, description])

After validating that the Incident ID doesn't already exist and that no fields were left blank, the function packages the incident_id, bot, and description into a small list and uses .append() to add it as a new record at the end of the incidents list.

Display Incidents:

for ticket in incidents:

```
print(f"{ticket[0]:<15} {ticket[1]:<20} {ticket[2]}")
```

This loops through every record currently stored in incidents and prints each one's ID, bot name, and description in neatly aligned columns, giving the user a full table view of all active tickets.

Search Incident:

for ticket in incidents:



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if ticket[0] == incident_id:

The function loops through the list and compares each ticket's ID (ticket[0]) against the ID the user typed in. If a match is found, it prints that ticket's full details and exits the function immediately using return.

Remove Resolved Incident:

incidents.remove(ticket)

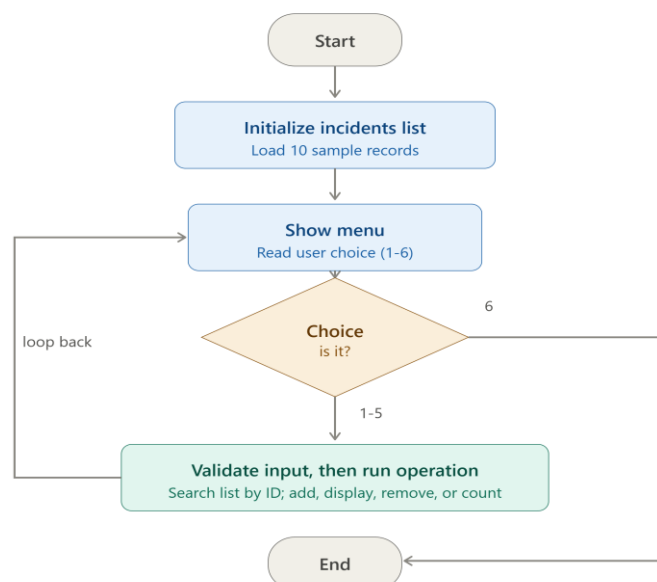
Similar to search, this loops through the list looking for a matching Incident ID. Once found, .remove() deletes that specific list entry from incidents, effectively closing out a resolved ticket.

Count Active Incidents:

print("Total active incident tickets:", len(incidents))

This simply uses the built-in len() function to count how many records are currently in the incidents list and prints that number.

III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.



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Through this activity, I learned how a simple linear data structure like a list can be used to build a working, real-world system such as an incident ticket tracker. The incidents list served as the backbone of the entire program, storing each ticket as a smaller list containing the incident ID, bot name, and description. By applying basic list operations like `.append()` to add new tickets, looping with `for` to display and search records, and `.remove()` to delete resolved ones, I saw how linear structures store data in a straightforward, ordered sequence that can be easily traversed from start to end. This activity helped me understand that even without complex data structures, a list is powerful enough to manage, organize, and manipulate real data efficiently, making it a fundamental tool in programming and data management.
